

Assignment 6 – Clustering

Description

We are going to implement clustering over both MySQL and MongoDB.

Your tasks

1.- Provide queries that prepare the data (points). The data must be represented as follows in MySQL:

Point(id, dim_1, dim_2, ..., dim_n)

And as follows in MongoDB:

Points: {_id: id, dim_1: v1, dim_2: v2, ..., dim_n: vn}

There will be two main sets of points as follows:

1.1.- For a given business category, businesses and their reviews. We will use the following attributes as dimensions: 1) latitude, 2) longitude, 3) restaurants_price_range, 4) total number of hours the business is open during weekdays (computed as the time difference between open and closing times), 5) total number of hours the business is open during weekends, 6) total number of categories, 7) count of reviews.

1.2.- For a given range of “yelping_since” dates, users and their total number of reviews. We will use the following attributes as dimensions: 1) useful, 2) funny, 3) cool, 4) total number of years as elite user, 5) total number of friends, 6) count of reviews, 7) average number of stars.

All dimensions must be DECIMAL(32, 24) in MySQL and Decimal128 in MongoDB. Check the grading software and query templates for more info.

2.- Provide queries that scale the dimensions of the points based on a given scaling method. In MySQL, use the squared z-score method to scale the data. Check the grading software and software templates for more info.

3.- Provide queries that, given points and centroids, compute distances between them (Manhattan or Euclidean). In MySQL, the expected result is as follows:

XYZ(pid, cid, distance)

where pid is the id of the point, cid is the id of the closest centroid, and distance is the distance between the centroid and the point.

In MongoDB, the expected result is as follows:

XYZ: {_id: pid, cid: c, distance: d}

Also, in MySQL, use the squared Euclidean distance. Check the grading software and software templates for more info.

4.- Provide queries that compute an assignment based on distances. Note: In MongoDB, you should create indexes for the pid field; otherwise, the performance is very slow. Check the grading software and software templates for more info.

5.- Provide queries that, given an assignment, compute the size of each cluster and the sum of squared errors of each cluster. Check the grading software and software templates for more info.

6.- Provide queries that, given an assignment and points, compute new centroids. Check the grading software and software templates for more info.

7.- Provide queries that compute the silhouette coefficient for a given set of points and assignment. For a point p_i , you must compute the mean distance to all the points assigned to the same cluster as p_i (including p_i as well), and the mean distances to all the points assigned to other clusters. Check the grading software and software templates for more info.

8.- Provide queries that compute mutual information for two given assignments. The output must be the probability of belonging to every other pair of clusters in the two assignments. Check the grading software and software templates for more info.

9.- For a given assignment and points, provide queries that implement several linkage methods to determine the distance among clusters. Check the grading software and software templates for more info.

Submission instructions

- Use the software template(s) and grading software provided in myCourses.
- Submit a single ZIP file to myCourses that must be named as your RIT user, e.g., `errvcs.zip`. Do not include '@rit.edu.' The file must contain folders named as the folders provided in the software template(s) containing your queries, Gradle or Poetry project(s).
- Everything will be graded on a Linux machine, so you must always use the exact names provided in this document, software template(s) and grading software.

Grading rubric

- All test cases in the grading software must pass with some restrictions like main memory and total running time.